

Top 5 Things I Learned about Windows Administration

Dominic Yeakel

PNC Cohort

Instructor: Andry Rakotomalala

8/17/2026

1. User Account Control (UAC) and Privilege Management

UAC prevents processes from silently gaining administrative rights by prompting for consent or credentials before elevation. This limits the damage malware or a compromised session can do, since most day-to-day work runs with standard-user privileges even when logged in as an admin. This, along with following least-privilege principles, separating admin and standard accounts, and minimizing who holds admin rights, is effective in reducing the scope of a breach.

2. Group Policy Object (GPO)

GPOs let admins enforce consistent security and configuration settings across many machines from a central point (AD). This matters because manually configuring each workstation is error-prone and doesn't scale. GPOs ensure password policies, software restrictions, firewall rules, and UAC behavior are applied uniformly, reducing the attack surface.

3. BitLocker and TPM

BitLocker provides full-disk encryption, protecting data at rest if a device is lost or stolen. The TPM (Trusted Platform Module) is the hardware chip that securely stores the encryption keys and verifies system integrity at boot, so BitLocker can unlock automatically only if the boot chain hasn't been tampered with.

4. Kerberos

The default authentication protocol in Windows domains, using a ticket-based system so passwords aren't sent repeatedly over the network. A client authenticates once with a Key Distribution Center (KDC, typically a domain controller) and receives a ticket-granting ticket, which it then uses to request service tickets for individual resources. This matters because it reduces exposure of credentials, supports mutual authentication (both client and server verify each other), and underpins single sign-on across the domain. However, it is also a common attack target, so admins need to monitor ticket usage and enforce strong service account hygiene.

5. EFS (Encrypting File System)

EFS is a Windows feature that encrypts individual files and folders at the file-system level, tied to a user's certificate/key rather than the whole disk. It's useful for protecting specific sensitive data on shared or multi-user machines without encrypting everything, unlike BitLocker's full-disk approach. The key risk admins must manage is key recovery. If a user's certificate is lost or corrupted without a proper recovery agent or backup, the encrypted data becomes permanently inaccessible, so EFS deployments require careful key/certificate management and recovery planning.